PHISHING EMAIL DETECTION & AWARENESS SYSTEM

CYBER SECURITY REPORT

FUTURE INTERNS

PREPARED **BY KIMBERLY FARAI MAGO**

OVERVIEW

Phishing is a form of cyber attack in which attackers attempt to trick individuals into revealing sensitive information such as usernames, passwords, financial details, or company data. These attacks are typically delivered through email and are designed to appear legitimate by imitating trusted organizations or internal departments.

In modern organizations, phishing remains one of the most effective attack methods because it exploits human behavior rather than technical vulnerabilities. Attackers use techniques such as urgency, impersonation, and deception to manipulate users into clicking malicious links, downloading harmful attachments, or performing unauthorized actions.

As phishing attacks continue to evolve, including the use of QR codes and brand impersonation, it is critical for employees to be able to recognize and respond to these threats effectively.

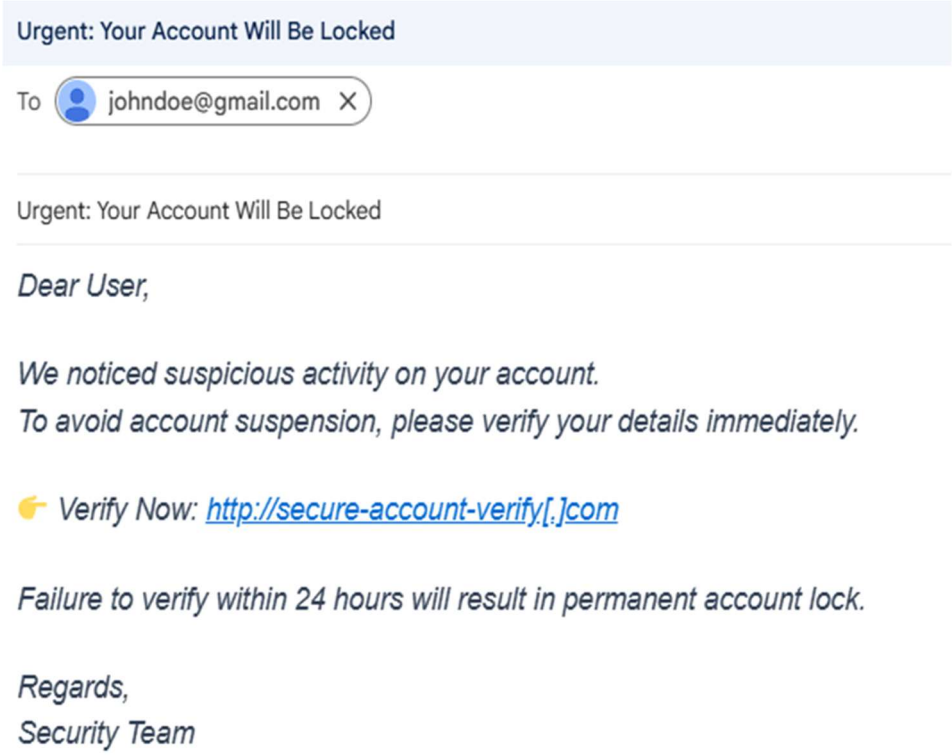
OBJECTIVE

The primary objective of this report is to analyze a range of phishing email examples in order to identify common attack patterns, techniques, and indicators used by cybercriminals. By examining these examples, the report aims to highlight how phishing attacks are structured and how they attempt to deceive users.

In addition, this report seeks to classify each analyzed email based on its level of risk, providing a clear understanding of how dangerous each type of phishing attempt can be. This classification helps in prioritizing responses and improving decision-making when handling suspicious emails.

Another key objective is to translate technical analysis into simple, user-friendly explanations that can be easily understood by non-technical individuals. Since phishing primarily targets everyday users, it is important that awareness materials are accessible and practical.

Ultimately, this report aims to support cybersecurity awareness by providing actionable guidance and best practices that individuals and organizations can use to detect, prevent, and respond to phishing attacks effectively.



Urgent: Your Account Will Be Locked

To johndoe@gmail.com

Urgent: Your Account Will Be Locked

Dear User,

We noticed suspicious activity on your account.
To avoid account suspension, please verify your details immediately.

👉 *Verify Now: http://secure-account-verify[.]com*

Failure to verify within 24 hours will result in permanent account lock.

Regards,
Security Team

METHODOLOGY

This report is based on the analysis of multiple phishing email samples obtained from publicly available phishing simulation platforms and security awareness resources. Each sample was carefully selected to represent different types of phishing attacks, including credential harvesting, malware delivery, brand impersonation, QR code phishing, and business email compromise.

The analysis process involved examining the content and structure of each email to identify key phishing indicators such as suspicious links, misleading domains, generic greetings, urgent language, and unusual requests. Special attention was given to how attackers use social engineering techniques to manipulate user behavior.

In addition to content analysis, commonly used cybersecurity tools and techniques were referenced to simulate real-world detection methods. These include email header analysis tools, browser-based link inspection, and domain verification practices. Although some technical elements were simulated, they reflect real processes used by security analysts.

Each email was then evaluated and classified based on its level of risk, taking into account the likelihood of user deception and the potential impact if the attack were successful.

GOALS

The primary goal of this analysis is to improve user awareness of phishing attacks by identifying common patterns and tactics used by attackers. The report aims to educate users on how to recognize and avoid potential threats.

PARTNERSHIPS

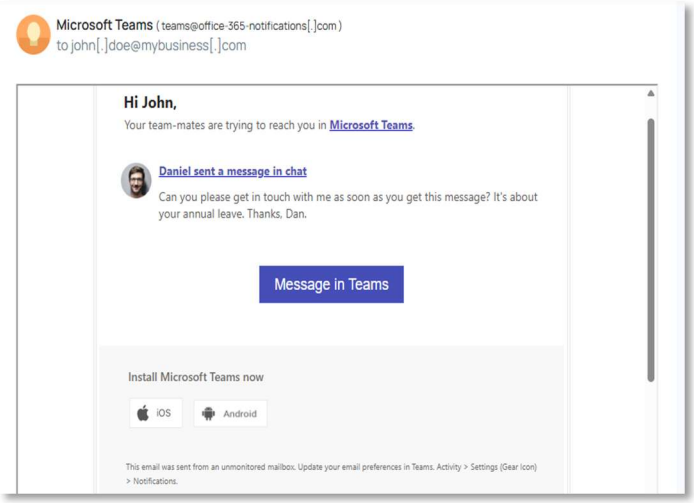
This report is based on industry practices and publicly available cybersecurity resources, including phishing simulation platforms and awareness training materials. These sources provide realistic examples of modern phishing attacks.

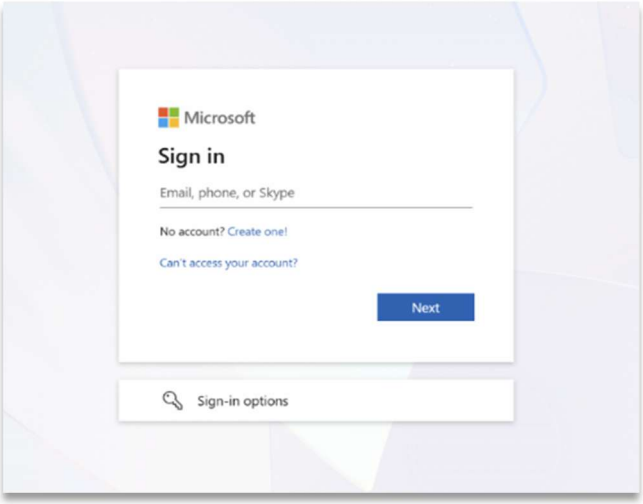
GROWTH

The findings from this analysis support continuous improvement in cybersecurity awareness. By understanding evolving phishing techniques, organizations can strengthen their defenses and reduce the risk of successful attacks.

PHISHING EMAIL ANALYSIS

SAMPLE 1





Threat Type: Brand Impersonation Phishing

Description:
This email impersonates Microsoft Teams by notifying the user of a missed message. It encourages the user to click a link to view the message, which redirects to a fake login page designed to steal credentials.

Phishing Indicators:

- Suspicious login link
- Email creates curiosity: “missed message.”
- No personalization
- Sender domain may not match Microsoft

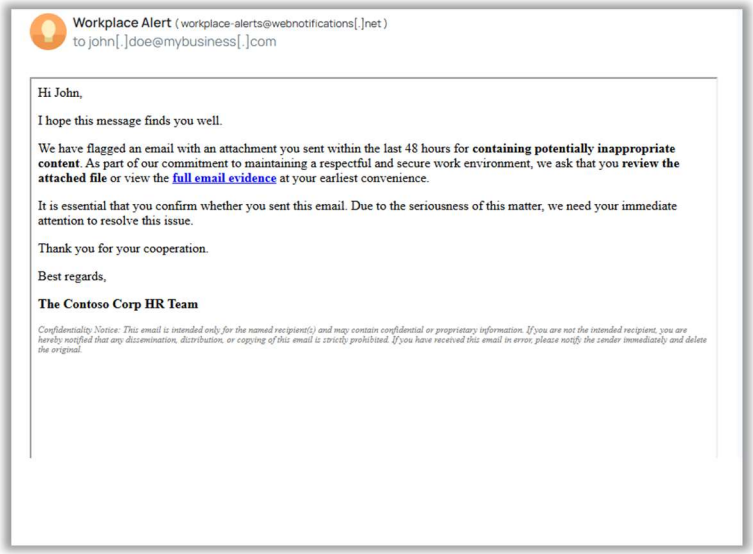
Risk Classification: Phishing (High Risk)

Detection Tools Used:

- Browser link inspection tools
- Email header analyzer (MXToolbox / Google Toolbox)
- Domain verification

How to Detect It:
Users should hover over the link to confirm it leads to an official Microsoft domain and avoid logging in through email links.

SAMPLE 2



Workplace Alert (workplace-alerts@webnotifications[.]net)
to john[.]doe@mybusiness[.]com

Hi John,

I hope this message finds you well.

We have flagged an email with an attachment you sent within the last 48 hours for **containing potentially inappropriate content**. As part of our commitment to maintaining a respectful and secure work environment, we ask that you **review the attached file** or view the **full email evidence** at your earliest convenience.

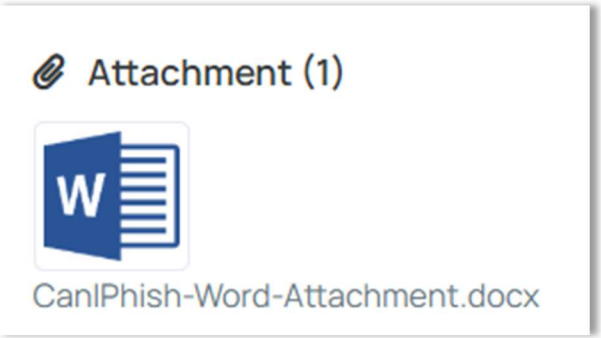
It is essential that you confirm whether you sent this email. Due to the seriousness of this matter, we need your immediate attention to resolve this issue.

Thank you for your cooperation.

Best regards,

The Contoso Corp HR Team

Confidentiality Notice: This email is intended only for the named recipient(s) and may contain confidential or proprietary information. If you are not the intended recipient, you are hereby notified that any dissemination, distribution, or copying of this email is strictly prohibited. If you have received this email in error, please notify the sender immediately and delete the original.



Threat Type: Malware Delivery Phishing

Description:
This email appears to come from the HR department and asks the employee to review an attached document. The attachment may contain malicious software designed to infect the system.

Phishing Indicators:

- Unexpected attachment
- Generic or unusual HR request
- Pressure to review document quickly
- Unknown or spoofed sender

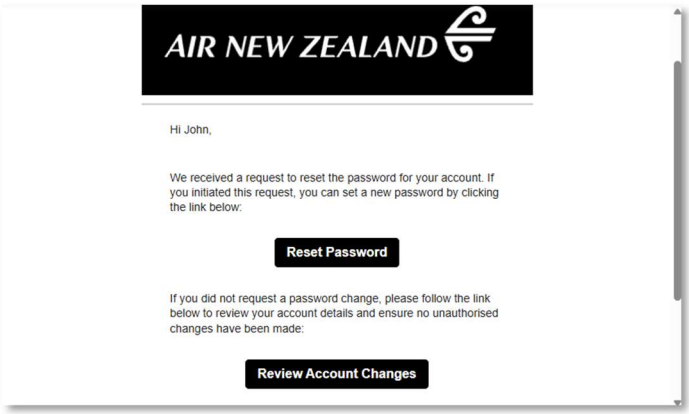
Risk Classification: Phishing (High Risk)

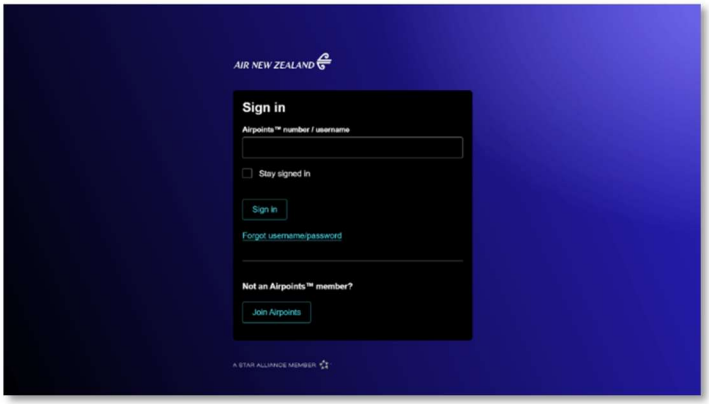
Detection Tools Used:

- Email attachment scanning tools
- Antivirus software
- Email filtering systems

How to Detect It:
Employees should verify HR-related emails directly with the HR department and avoid opening unexpected attachments.

SAMPLE 3





Threat Type: Credential Harvesting (Brand Impersonation)

Description:
This email impersonates Air New Zealand and claims that a password reset was requested. It directs users to a fake website to capture login credentials.

Phishing Indicators:

- Unrequested password reset
- Suspicious link
- Use of a well-known brand
- Urgent tone

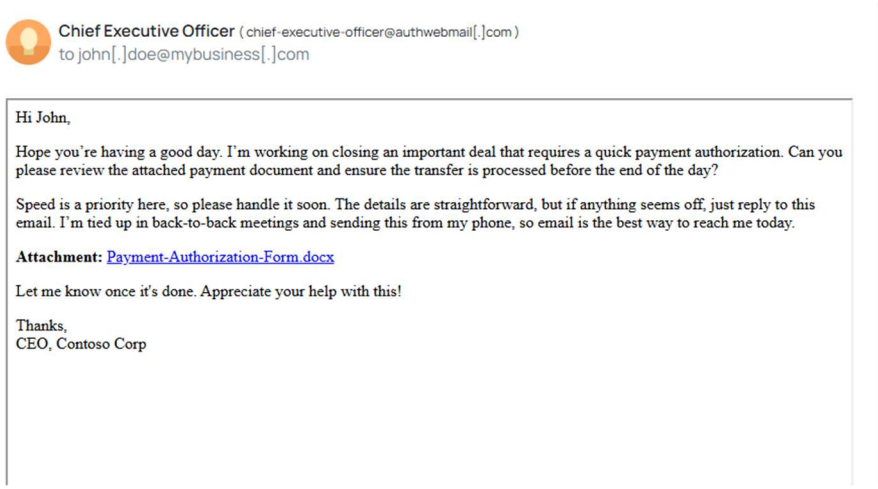
Risk Classification: Phishing (High Risk)

Detection Tools Used:

- Link inspection tools
- Email header analysis
- Domain verification

How to Detect It:
If no password reset was requested, the email should be ignored and reported. Users should only reset passwords through official websites.

SAMPLE 4



Chief Executive Officer (chief-executive-officer@authwebmail[.]com)
to john[.]doe@mybusiness[.]com

Hi John,

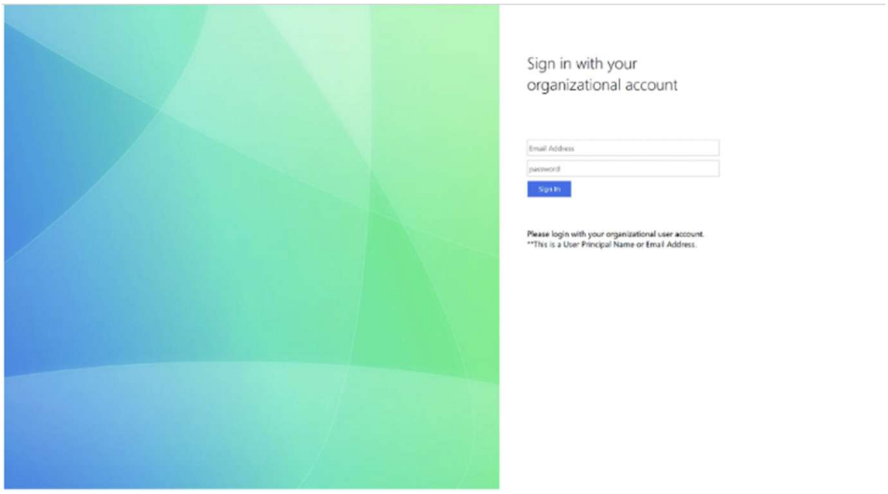
Hope you're having a good day. I'm working on closing an important deal that requires a quick payment authorization. Can you please review the attached payment document and ensure the transfer is processed before the end of the day?

Speed is a priority here, so please handle it soon. The details are straightforward, but if anything seems off, just reply to this email. I'm tied up in back-to-back meetings and sending this from my phone, so email is the best way to reach me today.

Attachment: Payment-Authorization-Form.docx

Let me know once it's done. Appreciate your help with this!

Thanks,
CEO, Contoso Corp



Threat Type: Business Email Compromise (BEC)

Description:
This email impersonates a company executive and requests an urgent financial transaction. It relies on authority and urgency to manipulate the employee.

Phishing Indicators:

- Urgent request for money
- Authority impersonation (CEO/Manager)
- Unusual or out-of-process request
- Lack of formal structure

Risk Classification: Phishing (Critical Risk)

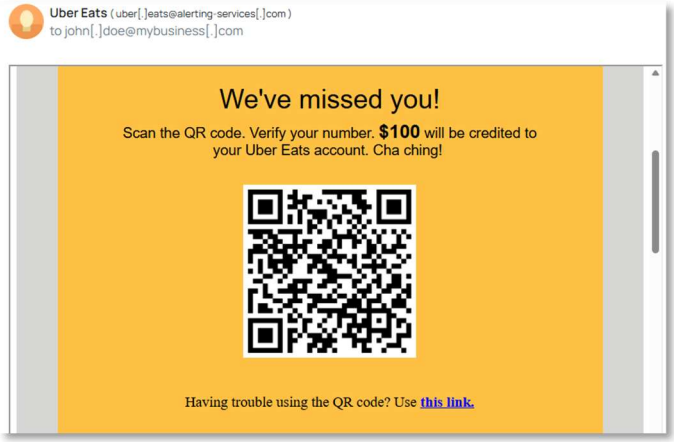
Detection Tools Used:

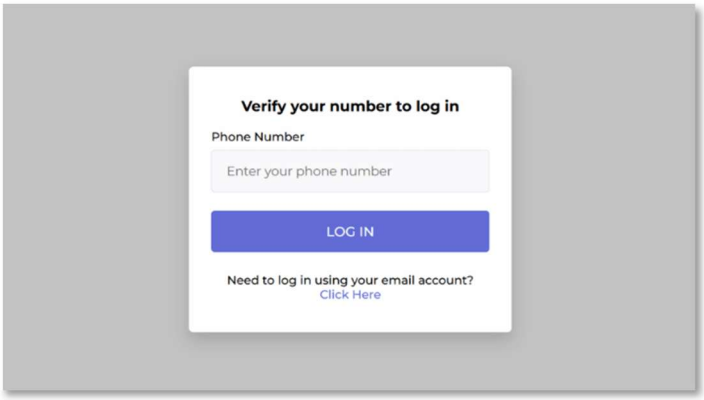
- Email header analyzer

- Internal verification procedures
- Security awareness practices

How to Detect It:
Employees should always confirm financial requests through official communication channels before taking action.

SAMPLE 5





Threat Type: QR Code Phishing (Quishing)

Description:
This email uses a QR code to trick users into scanning it with their mobile device. The QR code redirects the user to a malicious website designed to steal login credentials or install malware. QR phishing is increasingly used to bypass traditional email security filters, as QR codes can hide malicious links from detection systems.

Phishing Indicators:

- Presence of a QR code instead of a clickable link
- Urgent request to scan the code
- No clear or verified sender identity
- Instructions to bypass normal login methods

Risk Classification: Phishing (High Risk)

Detection Tools Used:

- Mobile browser link inspection
- QR code scanner preview (to reveal URL before opening)
- Email header analyzer (MXToolbox / Google Toolbox)

How to Detect It:
Users should avoid scanning QR codes from unknown emails. If necessary, they should preview the URL before opening it and verify the request through official channels.

PHISHING INDICATORS

Throughout the analysis of the phishing email samples, several common indicators were consistently observed. These indicators serve as warning signs that an email may be malicious and should be treated with caution.

One of the most prominent indicators is the use of urgent or threatening language, such as warnings about account suspension or immediate action required. This tactic is designed to pressure users into reacting quickly without verifying the authenticity of the message.

Another common indicator is the presence of suspicious or mismatched URLs. Attackers often create domains that closely resemble legitimate ones to deceive users. These links may lead to fake login pages or malicious websites.

Generic greetings, such as “Dear User” or “Dear Customer,” are also frequently used in phishing emails. Legitimate organizations typically personalize their communications, making generic greetings a potential red flag.

Additional indicators include unexpected attachments, impersonation of trusted brands or individuals, requests for sensitive information, and unusual formatting or grammatical errors. While not all phishing emails contain obvious mistakes, a combination of these indicators significantly increases the likelihood that an email is malicious.

In short, the common indicators that were identified across the analyzed phishing emails:

- Use of urgent or threatening language
- Suspicious or mismatched URLs
- Generic greetings, e.g., “Dear User.”
- Unexpected attachments or requests
- Impersonation of trusted brands or internal staff
- Requests for sensitive information (passwords, payments, OTPs)
- Poor or unusual email formatting

These indicators are key warning signs that users should look out for when evaluating emails.

RISK CLASSIFICATION

In order to assess the severity of each phishing attempt, the analyzed emails were categorized into different risk levels based on their characteristics and potential impact.

Emails classified as **Safe** show no signs of malicious intent and come from verified sources with legitimate content. These emails do not pose a threat to users or systems.

Suspicious emails contain some unusual or questionable elements, such as unexpected requests or minor inconsistencies. While they may not be immediately harmful, they require further verification before any action is taken.

Emails classified as **Phishing** present clear indicators of malicious intent. These emails often combine multiple attack techniques, such as spoofed sender addresses, deceptive links, and social engineering tactics. They pose a high risk to users and organizations.

In this report, the majority of analyzed samples fall under the **Phishing (High Risk)** category, highlighting the serious nature of these attacks and the importance of user vigilance.

The majority of the analyzed emails were classified as **Phishing (High Risk)** due to the presence of multiple attack indicators and the potential for serious impact.

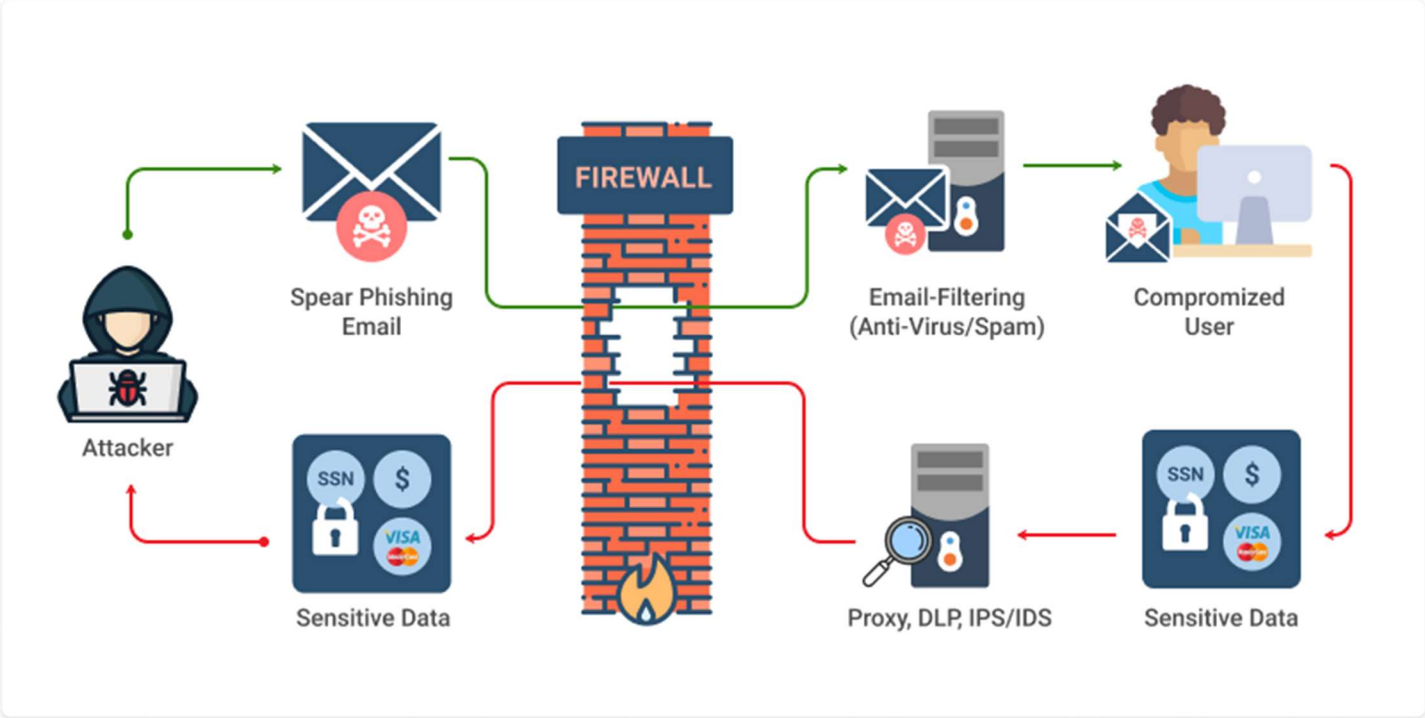
HOW THE ATTACK WORKS

Phishing attacks typically follow a structured and strategic process designed to maximize the chances of success. The attack begins with the creation of a convincing email that mimics a legitimate source, such as a trusted company, service provider, or internal authority figure.

The attacker carefully crafts the message to include elements that will trigger an emotional response, such as urgency, fear, or curiosity. For example, users may be warned about account suspension, informed of a missed message, or asked to complete an urgent task.

Once the user engages with the email, they are directed to perform an action. This may involve clicking on a malicious link, scanning a QR code, downloading an attachment, or responding with sensitive information. These actions often lead to credential harvesting websites, malware downloads, or fraudulent transactions.

After obtaining the desired information or access, the attacker can exploit it for various malicious purposes, including unauthorized account access, financial fraud, identity theft, or further attacks within the organization.



PREVENTION & AWARENESS GUIDELINES

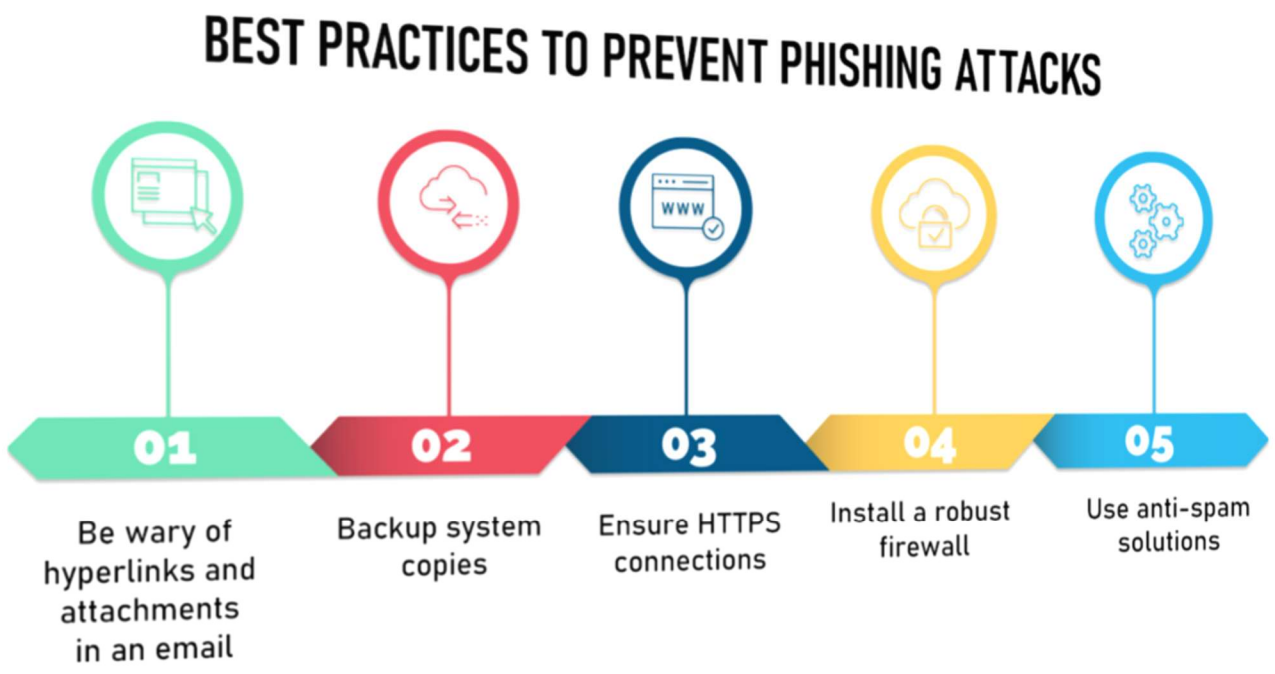
Preventing phishing attacks requires a combination of user awareness, cautious behavior, and the use of appropriate security tools. Since phishing primarily targets human decision-making, educating users is one of the most effective defense mechanisms.

Users should always verify the authenticity of emails before taking any action. This includes carefully checking the sender's email address, inspecting links before clicking, and being cautious of any unexpected or urgent requests.

It is also important to avoid downloading attachments from unknown or untrusted sources, as these may contain malware. Sensitive information such as passwords, banking details, or one-time passwords (OTPs) should never be shared via email.

Organizations should encourage employees to report suspicious emails to their IT or security teams. Prompt reporting allows potential threats to be investigated and mitigated before they can cause harm.

Regular cybersecurity awareness training is essential in keeping users informed about new and evolving phishing techniques, including modern threats such as QR code phishing.



DO'S AND DON'TS

DO:

- **Verify the sender's identity before taking any action**
  Always check the sender's email address carefully. Look for small changes in domain names or unusual formatting that may indicate spoofing or impersonation.
- **Hover over links to inspect the destination URL**
  Before clicking any link, place your cursor over it to preview the actual URL. Ensure that it matches the official website of the organization it claims to represent.
- **Use official channels to confirm requests**
  If an email requests sensitive information, payments, or urgent action, verify the request through trusted communication channels such as phone calls or official company platforms.
- **Report suspicious emails immediately**
  Inform your organization's IT or security team if you receive a suspicious email. Early reporting can help prevent potential attacks from spreading.
- **Be cautious of urgency and emotional triggers**
  Take a moment to think before acting on emails that create a sense of panic, urgency, or pressure. Attackers often rely on emotional manipulation to bypass logical thinking.
- **Keep security software and systems updated**
  Ensure that antivirus software, email filters, and operating systems are regularly updated to help detect and block known threats.

DON'T:

- **Do not click on links from unknown or untrusted sources**
  Clicking on malicious links can lead to phishing websites or trigger malware downloads that compromise your system.
- **Do not scan QR codes from unsolicited emails**
  QR codes can hide malicious links that bypass traditional detection systems. Avoid scanning them unless the source is verified.
- **Do not download or open unexpected attachments**
  Attachments from unknown or suspicious senders may contain harmful software such as viruses, ransomware, or spyware.
- **Do not share sensitive information via email**
  Legitimate organizations will never ask for passwords, OTPs, or financial details through email communication.
- **Do not respond to suspicious emails**
  Replying to phishing emails can confirm to attackers that your email address is active, potentially leading to more targeted attacks.
- **Do not ignore warning signs**
  Even if an email looks professional, small inconsistencies such as unusual requests, generic greetings, or mismatched links should not be overlooked.

CONCLUSION

Phishing continues to be one of the most prevalent and dangerous cybersecurity threats affecting individuals and organizations worldwide. This report has demonstrated how attackers use a variety of techniques, including social engineering, brand impersonation, malicious attachments, and QR code-based attacks, to deceive users.

The analysis of multiple phishing email samples highlights the importance of recognizing common indicators and understanding how these attacks are structured. By identifying warning signs such as suspicious links, urgent language, and unexpected requests, users can significantly reduce the likelihood of falling victim to phishing attempts.

Ultimately, the most effective defense against phishing lies in a combination of user awareness, critical thinking, and adherence to security best practices. Organizations that invest in continuous education and proactive security measures are better equipped to protect themselves against these evolving threats.